

Problem 279: Running on Fumes

Difficulty: Easy

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Problem Background

You're planning a family road trip and you want to make sure you don't get stranded without gas in the middle of nowhere. You need to know whether the gasoline left in the tank of your car will be enough to finish a planned route. Let's write a program that will help you reach your destination with enough gas and no headaches.

Problem Description

The fuel efficiency of a car's engine varies depending on three factors:

- **Cylinder count** – More powerful engines have more cylinders, and therefore burn more fuel.
- **Road type** – Cars are more efficient on highways where they can reach cruising speeds and have more consistent acceleration, as compared to city or off-road driving.
- **Fuel-level effect** – As the tank empties, the fuel vaporizes more easily, and the engine ends up consuming more fuel.

The car rental company you're using has three types of car available, and has provided you with information about each car's performance. They have a small sedan with a 4-cylinder engine, an SUV with a 6-cylinder engine, and a sport car with a powerful 8-cylinder engine. The base miles-per-gallon (MPG) for each of these cars is:

Cylinder Count	City MPG	Highway MPG	Off-road MPG
4	28	35	20
6	22	28	15
8	18	22	12

For each test case, you'll be given information about the car you're using and the route you're planning to take. You'll need to calculate the car's **effective MPG** for each segment of the route, and determine if you have enough fuel to finish the entire route. The **effective MPG** for a segment of the route can be calculated using this formula:

$$\text{Effective MPG} = \text{Base MPG} - (0.25 * (\text{Max Fuel} - \text{Current Fuel}))$$

"Current Fuel" will be the amount of gas in the tank at the start of the segment of the route. After each segment, you'll need to determine the amount of fuel left in the tank and use that to calculate the

effective MPG for the next segment. You cannot start a new segment if the remaining fuel is not greater than zero gallons.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include:

- A line containing four values separated by spaces:
 - A number 4, 6, or 8, representing the number of cylinders in the car's engine
 - A positive decimal representing the amount of gas in the car's tank at the start of the route, in gallons
 - A positive integer representing the maximum capacity of the car's gas tank, in gallons
 - A positive integer, **R**, representing the number of segments in the route
- **R** lines, each representing a route segment in sequence. Each line contains the following values separated by spaces:
 - A letter representing the type of road being used on this segment; C represents city roads, H represents highways, and O represents an off-road segment
 - A positive integer representing the length of the segment in miles

```
3
4 10.0 15 5
C 50
H 30
C 20
O 40
C 10
6 5.5 12 3
H 100
H 150
H 80
8 2.0 8 4
C 20
C 30
C 25
C 35
```

Sample Output

For each test case, your program must print a single line containing the uppercase word YES if all segments can be successfully completed without running out of gas, or the uppercase word NO otherwise.

YES

NO

NO